

Comparative Analysis of Modeling Methods for Local House Price Index Forecasting

By Raymond Ding

Predicting the All-Transactions House
Price Index for
Washington-Arlington-Alexandria,
DC-VA-MD-WV (MSAD)



Agenda

April 2025



01 Objectives

02 Methodology

03 Key Findings

04 Limitations

05 What I Learned

Project Objectives

April 2025

- **Compare Model Accuracy:** Evaluate various time series and machine learning models.
- **Identify Best Model:** Determine the top-performing model using walk-forward validation and Relative Mean Squared Error against a historical mean benchmark over an 8-quarter horizon.
- **Generate Forecast:** Produce an 8-quarter forecast (through Q4 2026) using the best model identified.

Methodology - Data Collection

April 2025

- **Target Variable:** All-Transactions House Price Index for Washington-Arlington-Alexandria, DC-VA-MD-WV (MSAD)
- **Frequency:** Quarterly
- **Time Period:** 1990 Q1 - 2024 Q4

- **Other Variables:**

- Local: Unemployment Rate(BLS), CPI(BLS), Median Household Income(Census Bureau), Population(Census Bureau)
- Federal: GDP(BEA), Interest Rate(FRED), Mortgage Rate(Freddie Mac), Poverty Rate(Census Bureau)

Preprocessing & Missing Values

April 2025

- **Challenge:** Some data was annual, or hasn't been updated since 2023
- **Strategies:**
 - Resampling by converting all data to Quarterly Start
 - Linear Interpolation
 - For values missing at the tail end: AutoARIMA

Preprocessing & Missing Values (Cont.)

April 2025

- **House Index (Quarterly):** Linear interpolation for any minor historical gaps.
- **CPI (Monthly):** Resampled to quarterly; linear interpolation for any gaps.
- **Poverty Rate (Annual):** Resampled to quarterly; linear interpolation for historical gaps; AutoARIMA for recent gaps (2023-Q1 to 2024-Q4) and future forecasting.
- **Median Household Income (Annual):** Resampled to quarterly; linear interpolation for historical gaps; AutoARIMA for recent gaps (2023-Q2 to 2024-Q4) and future forecasting.
- **Population (Quarterly):** Linear interpolation for any historical gaps; 1990-Q1 estimated via backward extrapolation; AutoARIMA for future forecasting.

Preprocessing & Missing Values (Cont.)

April 2025

- **Interest Rate (Monthly):** Resampled to quarterly; linear interpolation. Future point (2025-Q2) manually added based on trend.
- **Mortgage Rate (Weekly):** Resampled to monthly mean, then merged quarterly; linear interpolation for any gaps.
- **GDP (Quarterly):** Linear interpolation for any minor historical gaps.
- **Unemployment Rate (Monthly):** Resampled to quarterly; linear interpolation. Future point (2025-Q2) manually added based on trend.

- **Benchmark:** Historical means forecasting
 - Taking historical average percentage change and applying it to the last value, then multi-outputting steps 1-8
- **Training/Testing Range:**
 - Training: 9 Folds using Time Series Split from 1990-2015
 - Testing: 2016 Q1 to 2024 Q4
- **Error Metrics:**
 - $\text{Relative MSE} = \text{Model MSE} / \text{Benchmark MSE}$

Methodology– Walk-Forward Validation

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- **Process:** Iteratively retrain on expanding window, forecast 8 steps ahead

Example: ARIMA on House_Index

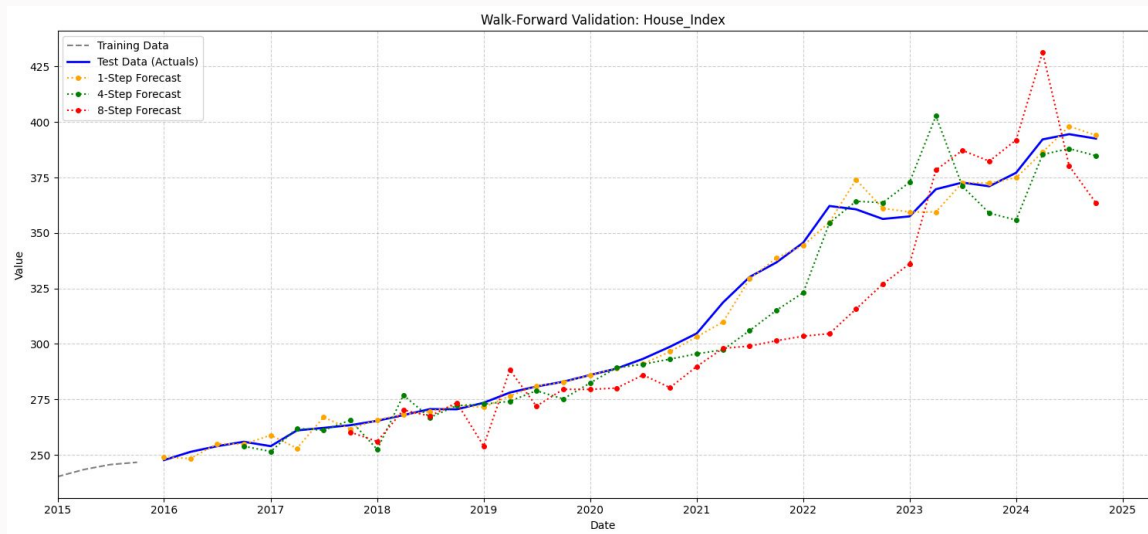


Figure 1: ARIMA's Walk-Forward Validation House_Index

AutoARIMA Results

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- **Unemployment_Rate:** ARIMA(2,0,1)(1,0,0)[4] -> d=0, D=0 (Suggests original series is stationary)
- **Interest_Rate:** ARIMA(2,1,2)(0,0,0)[4] -> d=1, D=0 (Suggests 1st order differencing needed)
- **Mortgage_Rate:** ARIMA(0,1,0)(0,0,0)[4] -> d=1, D=0 (Suggests 1st order differencing needed)
- **CPI:** ARIMA(1,1,0)(1,0,1)[4] -> d=1, D=0 (Suggests 1st order differencing needed)
- **House_Index:** ARIMA(0,1,3)(1,0,0)[4] -> d=1, D=0 (Suggests 1st order differencing needed)
- **GDP:** ARIMA(0,2,1)(0,0,0)[4] -> d=2, D=0 (Suggests 2nd order differencing needed)
- **Population:** ARIMA(1,1,0)(0,0,1)[4] -> d=1, D=0 (Suggests 1st order differencing needed)
- **Poverty_Rate:** ARIMA(2,1,0)(1,0,2)[4] -> d=1, D=0 (Suggests 1st order differencing needed)
- **Median_Household_Income:** ARIMA(1,1,3)(0,0,0)[4] -> d=1, D=0 (Suggests 1st order differencing needed)

AutoARIMA Forecast

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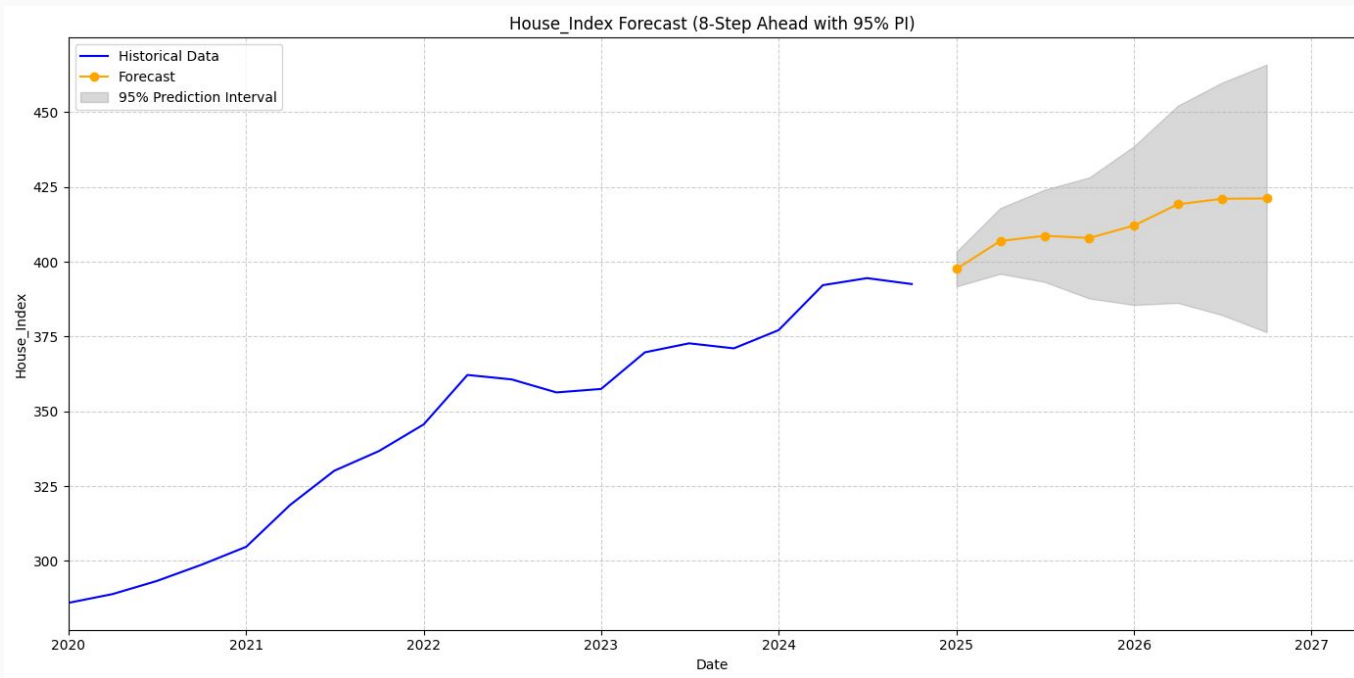


Figure 2: Final Forecast Using AutoArima

Why Did VECM Fail?

April 2025



Figure 3: VECM Model Residuals, after differencing

Why Did VAR Fail?

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- Switched to Percent Change differences; residuals were still too much

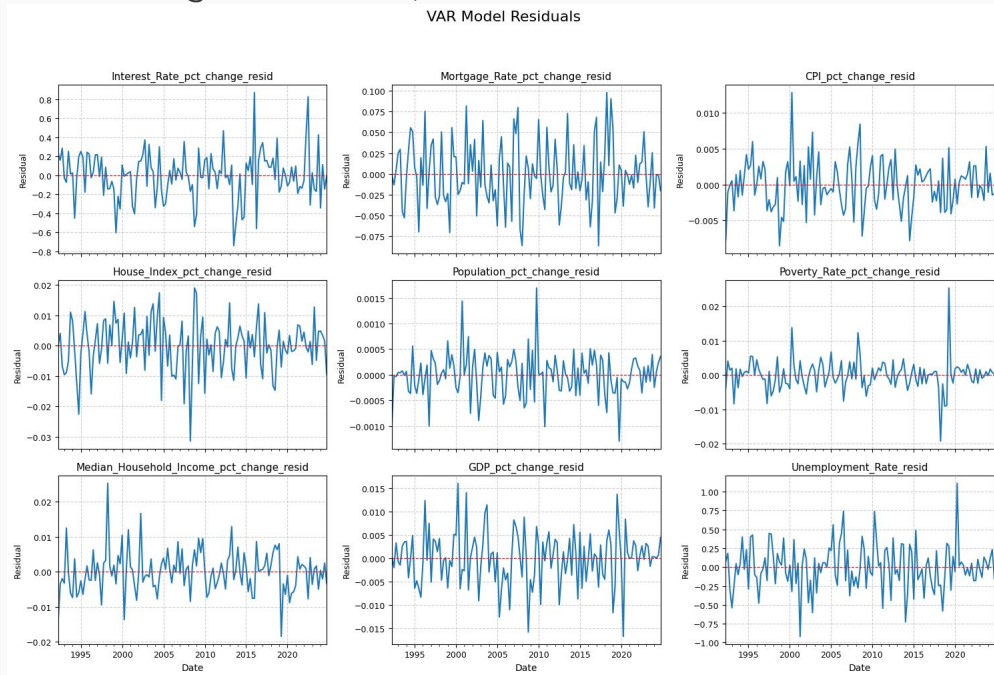


Figure 4: VAR Model Residuals, after pct change

Granger and Feature Engineering

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- Performed Granger Causality tests, only Unemployment rate at lag 4 was significant at p-value of 0.1

Unemployment_Rate -> House_Index_pct_change:

Lag 1: 0.422 | Lag 2: 0.704 | Lag 3: 0.234 | Lag 4: 0.100* | Lag 5: 0.184 | Lag 6: 0.302 | Lag 7: 0.309 | Lag 8: 0.408

- Create features for Regression/ML models (Lasso, ElasticNet, SVR, LSTM)
 - Lags (1-8): For target (% change) and all exogenous variables (% change/level).
 - Dummy Variables:
 - dummy_crisis_2008_period (2007 Q4 - 2009 Q4)
 - dummy_covid_2020_period (2020 Q1 - 2021 Q4)

PACF and ACF Visualization

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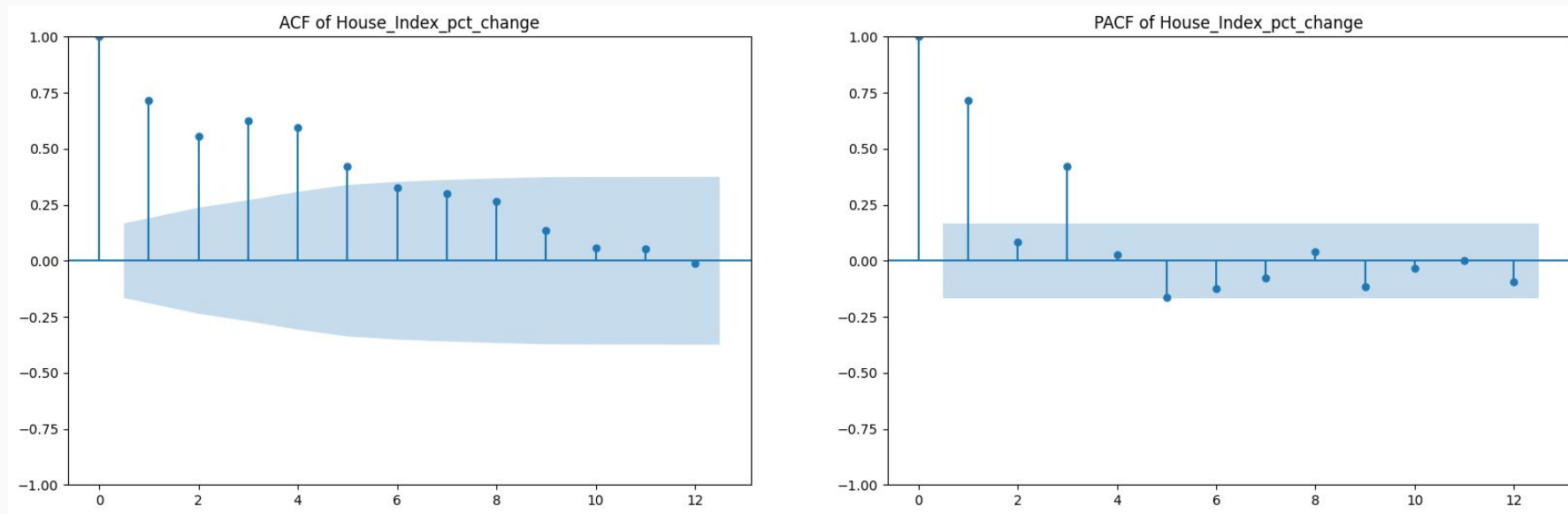


Figure 5: ACF and PACF plot showing autocorrelation on percent change on House Index

- Lasso and ElasticNet performs well for feature selection via L2 Regularization
 - ElasticNet includes a combination of L1 Regularization (Used in Ridge Regression) and L2 Regularization
- Hyperparameter tuning to update the new value was done for each step

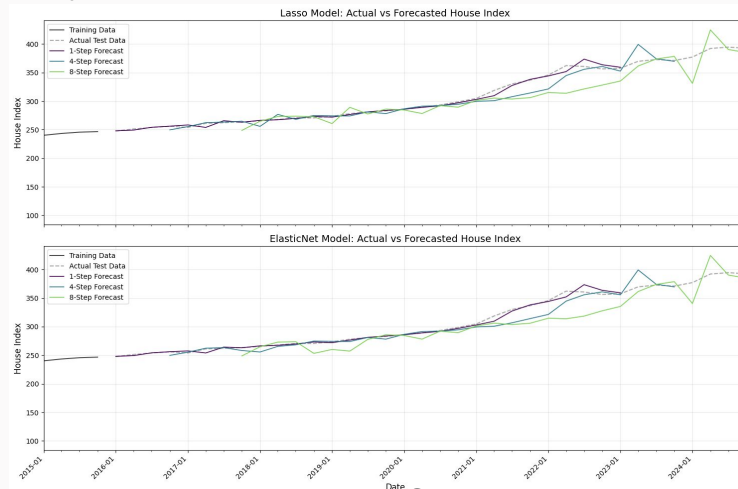


Figure 6: Walk-Forward Validation Results on Steps 1,4 and 8 on Lasso/ElasticNet

Lasso and ElasticNet Feature Selection

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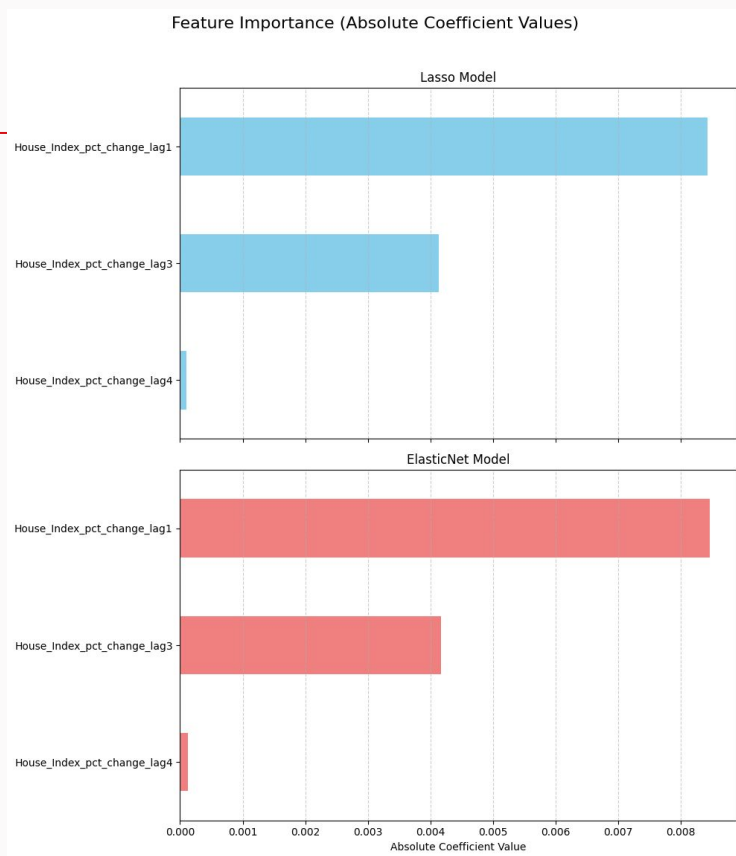


Figure 7: Feature Importance Ranked by Lasso and ElasticNet Regression

Support Vector Regression (SVR)

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- Support Regression Regression captures non-linear relationships very well
 - Single-Tuned

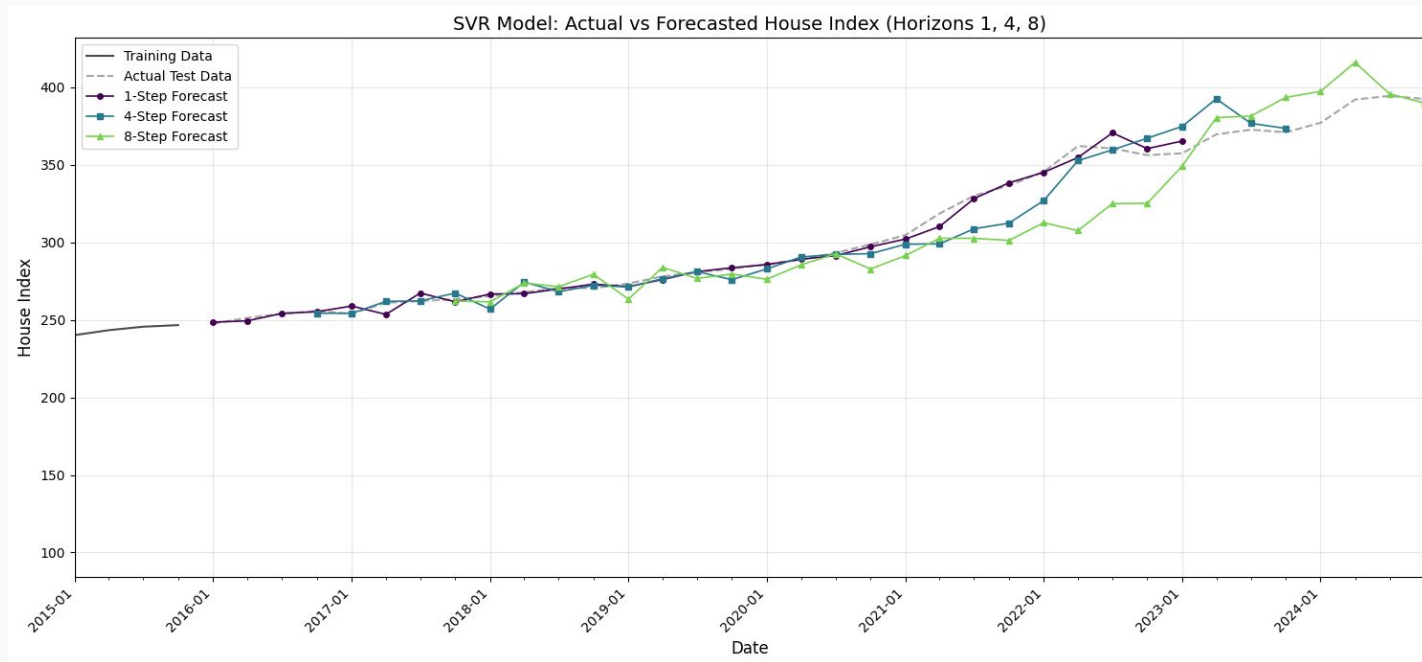


Figure 8: Walk-Forward Validation Results on Steps 1,4 and 8 on SVR

Long-Short-Term-Memory (LSTM)

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- Deep learning model for sequential data.
 - Multi-Tuned

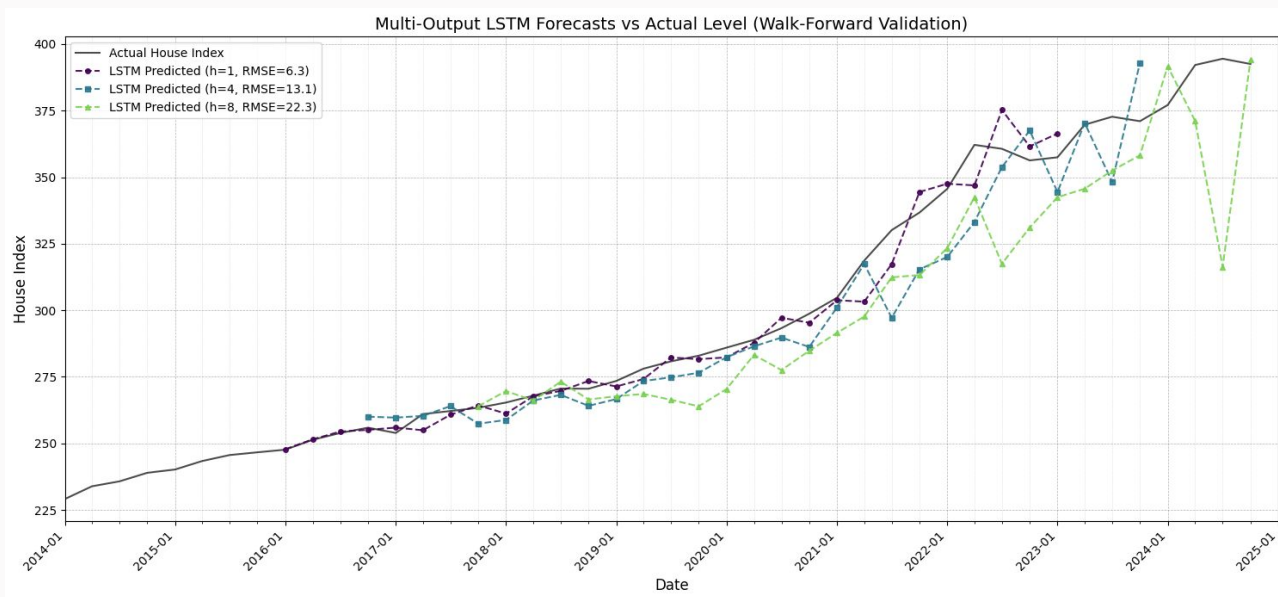


Figure 9: Walk-Forward Validation Results on Steps 1,4 and 8 on LSTM

DC House Index on 2024-10-01: 392.54

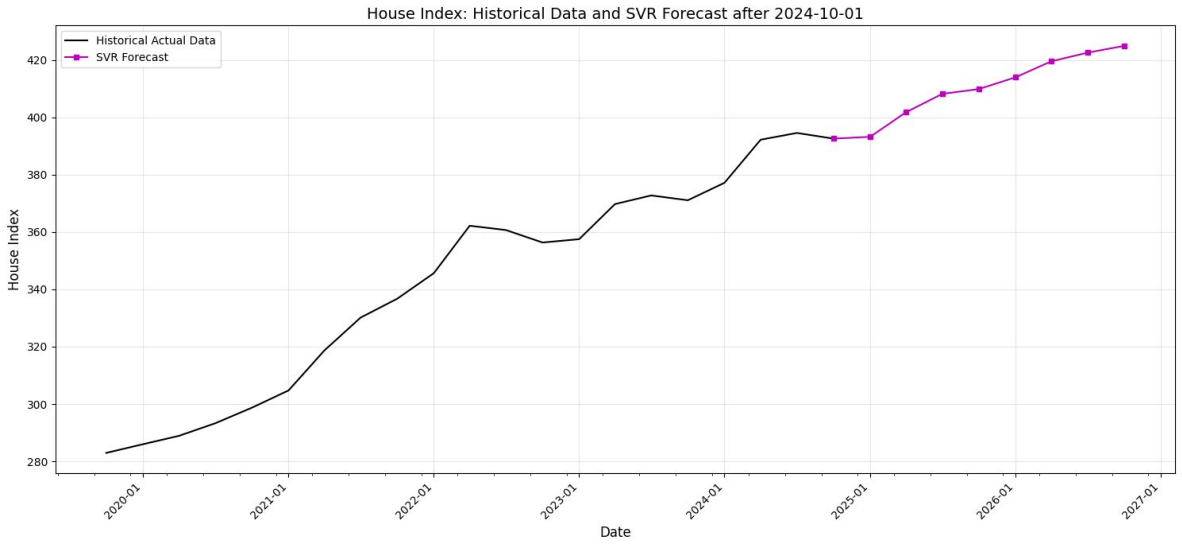


Table 1: Final SVR Forecast Values (Rounded)

Date	SVR Forecast
2025-01-01	393.17
2025-04-01	401.74
2025-07-01	408.14
2025-10-01	409.82
2026-01-01	413.92
2026-04-01	419.50
2026-07-01	422.51
2026-10-01	424.88

Figure 10: Final Forecast Using Highest Performing Model (SVR)

Table 2: Model Performance Ranking (Relative MSE) Across Prediction Steps

Steps	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5
1	SVR (0.791)	LSTM (0.874)	ElasticNet (0.886)	ARIMA (0.907)	Lasso (0.923)
2	SVR (0.676)	ARIMA (0.948)	ElasticNet (0.975)	Lasso (1.000)	LSTM (1.074)
3	SVR (0.768)	ElasticNet (0.955)	Lasso (0.970)	ARIMA (0.990)	LSTM (1.209)
4	SVR (0.774)	ElasticNet (0.867)	Lasso (0.873)	ARIMA (1.028)	LSTM (1.038)
5	SVR (0.858)	ElasticNet (0.994)	Lasso (1.007)	LSTM (1.031)	ARIMA (1.259)
6	SVR (0.966)	ElasticNet (1.095)	Lasso (1.112)	LSTM (1.166)	ARIMA (1.429)
7	SVR (1.013)	Lasso (1.093)	ElasticNet (1.097)	LSTM (1.450)	ARIMA (1.513)
8	SVR (1.000)	Lasso (1.087)	ElasticNet (1.092)	ARIMA (1.409)	LSTM (1.493)

Note: Relative MSE (shown parenthetically in Table 2) is calculated against a historical means benchmark model. This benchmark uses the average percentage change from historical data, is multi-outputting, updates the historical mean at each step, and follows the same walk-forward validation procedure as the tested models. A value < 1 indicates the model outperformed the benchmark; a value > 1 indicates the benchmark outperformed the model. Lower Relative MSE corresponds to a better rank.

- **Data Latency:** Real-world delays in economic data releases.
- **Structural Breaks:** Models may not fully capture recent, unique economic shifts (post-COVID, inflation); dummy variables are approximations.
- **Feature Influence:** Regularized regression limited the apparent impact of broader economic factors; non-linear models might capture more.
- **Econometric Models:** VECM/VAR implementation challenges due to residual issues; log-differencing not explored.
- **Tuning:** Potential for further improvement with more exhaustive hyperparameter optimization.

- **Log Differencing:** Revisit VECM/VAR and other models using log differences.
- **Formal Structural Break Testing:** Implement tests (Chow, Bai-Perron) and incorporate findings (e.g., regime-switching models).
- **Advanced Features/Models:** Explore interactions, non-linearities, alternative data, Gradient Boosting, Random Forests.
- **Ensemble Methods:** Combine top model forecasts.
- **Exogenous Forecasts:** Improve methods for predicting future inputs (beyond AutoARIMA).
- **LSTM Refinement:** Deeper tuning, explore GRU/Attention, fix plotting code.

- **Data Handling:** Importance of managing diverse time series data (frequencies, missing values, interpolation vs. model imputation).
- **Stationarity:** Practical application of tests (ADF/KPSS) and transformations (differencing, % change).
- **Evaluation Rigor:** Value of walk-forward validation and benchmarking for time series.
- **Model Suitability:** Understanding practical challenges (VECM/VAR residuals) and relative strengths (SVR simplicity/effectiveness, LSTM complexity).
- **Tuning Techniques:** Exposure to various hyperparameter optimization methods.
- **Iterative Process:** Forecasting is rarely linear; requires adapting based on results.

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- **Objective Achieved:** Successfully compared multiple forecasting models for the DC House Index.
 - **Best Model:** SVR (using lags 1-3) provided the most accurate forecasts relative to the benchmark.
 - **Forecast Generated:** A forecast through Q4 2026 was produced using the SVR model, predicting continued growth.
 - **Insights Gained:** Highlighted the strong autoregressive nature of the house index and the challenges of traditional econometric models on this dataset.

- George Milunovich. Forecasting australia's real house price index: A comparison of time series and machine learning methods. Journal of Forecasting, 39(7): 1098–1118, 2020.
- Meszaros, John. “A Brief Review of House Price Forecasting Methods.” Counselors of Real Estate, February 7, 2024.
<https://cre.org/real-estate-issues/a-brief-review-of-house-price-forecasting-methods/>.